

Ahmer Nadeem Khan

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PhD student in Financial Mathematics working in high-performance computing and quantitative finance, with a research focus on **limit order book simulation**, **portfolio optimization**, and **Monte Carlo**. Experienced in building, testing, and communicating computational and mathematical research ideas across academic research teams.

Education

Florida State University – PhD in Financial Mathematics Sep 2024 – May 2029

- Current GPA: 3.93, Supervisor: **Dr. Alec Kercheval**
- Coursework: Financial Engineering I/II, Monte Carlo Methods in Finance, Advanced Probability, Computational Math I/II, High Performance Computing, Differential Geometry, Measure Theory I/II, Numerical Optimization, Stochastic Analysis, Numerical Linear Algebra
- Current Coursework: Selected Unsolved Problems in Pure Mathematics

Lahore University of Management Sciences, Pakistan – BS in Mathematics Sep 2020 – May 2024

- Minor in Computer Science, Minor in English Literature
- GPA: 3.84
- Graduate Coursework: Machine Learning, PDEs, Topology, Algebraic Geometry, Algebraic Topology

Seoul National University, South Korea – Semester Abroad Sep 2023 – Dec 2023

- Department of Mathematical Sciences
- Coursework: Algebraic Topology II, Korean Literature
- Research thesis on Schubert Calculus for algebraic and arithmetic geometry under Dr. Jun Ho Whang

Research and Teaching Experience

Research Assistant, Department of Computer Science, Florida State University Jan 2026 – Present

- Collaborating with researchers at Argonne National Laboratory for scientific computing under **Dr. Kai Zhao**
- Developing warm-started randomized SVD for adaptive, error-bounded compression of streaming scientific data
- Benchmarked across three large-scale simulations (hurricane, cosmology, turbulence) spanning **16.7 billion** grid points, with a hard guarantee on maximum pointwise error
- Warm-start initialization cut storage up to **21%** and improved fidelity by **+2.2 dB** PSNR over the cold-start baseline, while running up to **3.4×** faster

Instructor of Record, Florida State University Jan 2026 – May 2026

- Sole instructor for MAC 2311 (Calculus with Analytic Geometry), delivering three weekly lectures (50 min) and one weekly recitation (75 min) to a class of 25 students
- Designed and delivered lecture materials and recitation activities, including problem sets and quiz reviews
- Authored, administered, and graded quizzes and examinations; handled proctoring, assessment, and student communication; managed course organization via Canvas

Graduate Teaching Assistant, Florida State University Sep 2024 – Dec 2025

- Assisted instructors across seven undergraduate courses including Trigonometry, Pre-Calculus, and Algebra
- Supported grading, proctoring, weekly labs, and organization of course materials

Teaching Assistant, Lahore University of Management Sciences Sep 2022 – May 2024

- Assisted instruction in Discrete Mathematics, Calculus III, and Linear Algebra
- Conducted weekly problem-solving sessions, tutorials, and office hours, integrating LaTeX and coding exercises
- Designed, proctored, and graded quizzes, homework, and examinations

8th Grade Teaching Volunteer, Syedanwala High School – Pakistan Jun 2023 – Aug 2023

- Led a team of six volunteers teaching students in rural Punjab
- Designed and delivered lessons in Algebra, Number Theory, and Geometry
- Conducted diagnostic tests, evaluations, and teacher training sessions for local staff

Selected Research Projects

Statistical Alpha Signal Research & Robust Backtesting for US Equities	June 2026
- Built a Python research platform and screened four alpha factors on 47 US large-caps (2020–2026) - Validated with overlap-adjusted t-tests, bootstrap, and robustness checks for dollar-neutral L/S portfolios - Best signal (12–1 momentum) delivered 5% annualized alpha vs. SPY ($\beta \approx 0$) at 15-day rebalance with 10 bps costs	
Dynamic VIX-Based Hedging of S&P 500 Risk under Tariff Shock Regimes	Nov 2025
- Selected as a top project from 107 projects in the Erdős Institute Fall 2025 Cohort - Built a regime-detection model using macro event triggers to hedge S&P 500 exposure via VIX futures ETF - Out-of-sample: reduced volatility by 47%, improved Sharpe ratio 2.6× and cut max drawdown from –19% to –4%	
Portfolio Optimization with James–Stein Shrinkage and Data Extraction Pipelines	May 2025
- Automated extraction of 1M+ data points of S&P 500 constituent and Treasury data (Yahoo Finance, U.S. Treasury) - Designed a walk-forward backtest with covariance shrinkage on cleaned price data for S&P 100 GMV portfolios - Shrinkage estimator reduced annualized volatility by 8.6% with a 2.8× Sharpe ratio improvement	
Exotic Option Pricing with CUDA-Accelerated Monte Carlo for HPC Applications	Apr 2025
- Priced Arithmetic Asian options, Lookback variants, and Barrier options using high-frequency simulations in C++ - Tested and optimized GPU implementation of Gaussian distribution sampling using parallel methods - GPU vs CPU: 5 million paths, 10,150× speedup factor, 96–98% accuracy retained	
Numerical and Monte Carlo Algorithms Library Development in MATLAB and Julia	Sep 2024 – May 2025
- Series of experimental projects focusing on algorithmic design, numerical stability, and convergence behavior - Implemented and benchmarked canonical algorithms from the numerical analysis and Monte Carlo literature - Developed a reproducible library of routines covering factorization and approximation methods, linear and non-linear system solvers, quadrature, variate-reduction, and PDE/ODE solvers	
Schubert Calculus and The Cohomology of the Grassmannian	Sep 2023 – May 2024
- Undergraduate research thesis conducted at Seoul National University (Grade: A+) - Explored algebraic and combinatorial techniques in intersection theory and enumerative geometry - Presented and defended thesis at Lahore University of Management Sciences	

Selected Talks and Presentations

Catalan Numbers and Their Properties , Algebra Seminar, Florida State University	Feb 2026
Presentation on the combinatorial descriptions of the Catalan numbers, and proof of various bijections and results	
Portfolio Optimization and Successive Convex Approximation , Florida State University	Dec 2025
Presentation on recent results on the application of the SCA framework to higher order (MVSK) portfolios	
Limit Order Book Simulation: A Review and Recent Progress , Financial Math Seminar, Florida State University	Dec 2025
Survey of point processes, agent-based models, deep learning, and generative AI for limit order book modeling	
Market Microstructure and High Frequency Trading , Financial Math Seminar, Florida State University	May 2025
Overview of HFT evolution, data structures, order-book mechanics, and algorithmic trading strategies	
Borsuk–Ulam Theorem and its Applications , Lahore University of Management Sciences	Dec 2024
Directed research (Topology) presentation on Tverberg’s Theorem and the Necklace Splitting Problem	

Honors and Awards

Khawaja Dil Muhammad Fellowship (sbasse.lums.edu.pk/honorific-fellowships)	2021 – 2024
Merit-based fellowship in Mathematics awarded for maintaining the highest departmental GPA	
Dean’s Honor List Awards	2021 – 2024
Annual university-wide recognition for academic excellence based on cumulative GPA	
Award of High Distinction	2024
Graduated with High Distinction, ranking 2 nd in BS Mathematics cohort	

Technical Skills & Certifications

Programming Languages & Platforms: Python (4 years), C++ (6 years), MATLAB (4 years), Julia (1 year), R (1 year), CUDA (1 year), LaTeX (6 years)

Libraries & Tools: pandas, NumPy, statsmodels, scikit-learn, Eigen3, BLAS, Git

Certifications: Erdős Institute Quant Finance Bootcamp, Akuna Options 101, Akuna Options 201